

Department of Computer Science**Assignment 2****Course Code: CS-151****Name:** _____**Course Title: Database Systems****Registration No.** _____**Total Marks: 20****Discipline/Semester: BSCS 2nd****Date: 04-04-25****Submission Date: 10-4-25****Instructions:**

- **Late submissions will not be accepted.**
- **Copied assignments from any source (including AI) will be marked zero.**

Q-1.**[CLO1, GA2, C2]**

Using the given scenario, **demonstrate** your understanding of relational database concepts and data modeling by addressing the following tasks related to entities, relationships, keys, constraints, and data models.

Part a:

The Central Library of Metropolitan University is planning to digitize and modernize its data infrastructure. The new system will include data on books, students, authors, and publishers. The objective is to implement a robust and scalable database system that can support various academic and administrative functions.

Using the given scenario:

1. Create relational tables for each entity involved in the system. Clearly define the table name, list of attributes, data types, and identify the primary key for each.
2. Propose a conceptual data model for the library system. Use high-level entities and highlight the key relationships between them (e.g., one-to-many, many-to-many).
3. Convert your conceptual model into a logical data model, showing the use of foreign keys and data types.
4. Discuss how a composite key might be used in a relationship.
5. Describe domain integrity, entity integrity, and referential integrity from this system.
6. Define candidate keys and alternate keys using the Book entity.
7. The company is considering different database models. Compare and contrast Relational, Object-based, and Semi-structured models in the context of this platform. Which model would you recommend and why?
8. Identify potential user-defined integrity constraints.